

1. General Introduction and Professional Context

1.1 - Professional Context and Stakeholder Roles

This project examines the production and use of cloud-cost information within the host organization's FinOps reporting process. The project framing describes a central Data Platform function, a Power BI team and a separate FinOps team. FinOps defines the financial questions and reviews the resulting indicators; the platform and reporting functions provide the technical means to prepare and present them. These responsibilities explain why a change to a cost measure can involve several teams even when the underlying data is already available.

The existing Cloud Showback dashboard serves FinOps, its management and managers in other domains, according to the student's account. The issue examined here is therefore the reliability and adaptability of the information supplied through that process, together with the degree of analytical autonomy available to authorized users. Section 1.2 describes the reported change workflow.

1.2 - Current Process and Diagnosis

The student reports that the current Power BI dashboard is used by FinOps, its management and managers of domains and subdomains in other teams. Dashboard access is therefore already part of the current process. The remaining question is whether each group can obtain the analyses it needs within its authorized scope and how changes to those analyses are made. Effective permissions and unmet requests have not yet been documented.

In the change workflow described by the student, a supervisor reviews the dashboard and provides feedback. The manager relays requested changes to an external contributor, who implements them. The student reports that these handoffs lengthen corrections. No request log, elapsed-time series or iteration count is currently available to quantify that effect. The contributor's location describes the service arrangement; it does not explain the delay by itself.

This account identifies a process dependency that can be tested. A request may require business approval, technical interpretation, implementation and acceptance, each with its own waiting time. The analysis in section 9.2 considers whether closer Data Engineering support within FinOps could reduce some handoffs while preserving review and publication controls. That proposal remains an organizational hypothesis until comparable requests and effort are measured.

1.3 - Problem Statement and Research Questions

The project addresses the transformation of detailed FOCUS cost and usage records into reliable data products. Those products must support several consumers while preserving shared definitions, access control, traceability and confidentiality. The reported dashboard-change workflow makes the way those products are maintained and used part of the problem.

The research question is:

How can a FinOps data platform transform FOCUS cost and usage data into reliable data products and provide governed self-service analytics to managers and application owners while supporting Inform, Optimize and Operate?

1.3.1 Three complementary subquestions

The central question is divided into three complementary subquestions rather than independent projects. They reflect the initial framing and organize the thesis argument.

1. Platform reliability: How can FOCUS cost and usage data become reliable, traceable and reusable data products while addressing security and scalability requirements?
2. Governed self-service: How can managers and application owners explore their authorized scope without manual FinOps extracts while preserving confidentiality and consistent indicators?
3. FinOps decision support: How can these products improve cost visibility, support analysis of optimization opportunities and enable recurring Inform, Optimize and Operate activities?

The first will be assessed through contract and quality controls, lineage, reconciliation and reproducibility, without conflating local execution with target-platform validation. The second requires user scenarios and access tests for every exposed channel; a usable interface alone does not establish secure self-service. The third requires correct and useful KPIs, separating cost differences, recommendations and measured savings.

1.3.2 Scope of self-service

The project aims for centralized FinOps data production and distributed consumption according to user permissions. Independent access does not require each domain to produce and maintain its own data. Section 4.2 develops this position and its relationship to Data Mesh; chapter 9 discusses its implications.

1.3.3 Additional design questions

Supporting questions address:

- the balance between Data Engineering depth, governance, cost and feasibility;

- the Data Contract and controls required to establish trust;
- the design of ownership, RLS and analyst access;
- the cost measures that support realized rate comparisons;
- the additional data required for potential-savings recommendations.

1.4 - Objectives, Scope and Constraints

The project aims to turn FOCUS cost-and-usage files into traceable analytical products. Its technical objectives are to validate a versioned data contract, preserve source lineage, reconcile charges across Bronze, Silver, Gold and datamarts, and make the resulting measures available to a reporting application. The evaluation also examines whether users can answer defined FinOps questions from these products and whether the platform records enough information to diagnose a failed or repeated run.

The empirical dataset begins with anonymized Azure FOCUS files and extends them synthetically. One local DuckDB and Streamlit prototype supports January 2025 validation. The active cloud implementation uses Databricks on GCP with separate DEV and PROD catalogs. These settings serve different tests; a local result cannot establish cloud behavior. The enterprise Azure architecture considered in the initial framing remains a design context rather than an implemented deployment.

Confidentiality prevents publication of the organization's detailed billing history. The limited reference data and synthetic extension constrain business conclusions, especially claims about savings or forecasting. Time, available cloud resources and incomplete access evidence constrain the performance and security evaluation. Section 3.4 defines the evaluation protocol; chapter 7 distinguishes completed tests from those still required.

1.5 - Summary of Requirements

The project framing groups ten acceptance requirements under platform reliability, governed self-service and FinOps decision support. Reliability requires identifiable historical and incremental batches, a versioned FOCUS contract, traceable Bronze-to-Gold transformations, reconciliation and repeatable execution. The data products must record their definitions, owner, freshness and quality before they are presented as certified within the project.

Governed self-service requires more than a working dashboard. The proposed acceptance scenario asks a manager to answer cost questions for an authorized domain and an analyst to produce an additional aggregation from a certified datamart. Access tests must cover the dashboard, exports and direct queries; filtering one interface does not secure another. These scenarios remain acceptance criteria until representative identities and permissions have been tested.

Decision support covers showback, period comparisons, documented cost formulas, pipeline status and recovery. The requirements distinguish a calculated price difference from a verified saving and require unknown allocation to remain visible. Chapter 7 reports what the prototype establishes against these criteria. The complete requirement matrix and its proposed evidence are retained in the project framing document.

1.6 - Thesis Structure

Chapter 2 defines the FinOps and cloud-cost concepts used in the analysis. Chapter 3 derives required artifacts from the intended outcomes, then explains dataset preparation, quality and governance, progressive implementation, analytical preparation and data-product delivery. It defines acceptance criteria without assuming that every proposed test has succeeded. Chapter 4 presents the logical and physical architecture, technology choices and analytical data model. Chapter 5 follows the three implementation workstreams: the independent FOCUS generator, the local DuckDB/Streamlit proof of concept, and the Databricks cloud platform. This ordering explains the final system; it is not a chronology of development. Chapter 6 interprets the available cost and allocation results. Chapter 7 assesses local validation and the evidence currently available for Databricks, including the limits of a performance comparison.

Chapter 8 examines build and operating costs and applies FinOps reasoning to the platform's own workloads, including the later workspace and compute decisions. Chapter 9 discusses the findings, limitations and organizational options. Chapter 10 returns to the research question and identifies work that depends on further evidence. Throughout the thesis, observed results are separated from reported execution and proposed tests.